PRODUCT USER MANUAL

For Global Ocean Reprocessed in-situ Observations of biogeochemical Product

INSITU_GLO_BGC_DISCRETE_MY_013_046
Issue: 3.0

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RECORD TABLE

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2.1	07/09/2020		New PUM format	V.S. Lien, V. Racapé	
2.2	31/08/2021		EMODnet chemistry inclusion	V.S. Lien, V. Racapé	S. Tarot
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2.6	01/06/2023		Inclusion of parameters SLCW and PHOW	V.S. Lien	S. Tarot
2.7	17/05/2024		New download description	V.S. Lien, J.E.Ø. Nilsen	S. Tarot
3.0	20/05/2025		Inclusion of Easy BGC dataset and Carbon dataset	E.M. Jones, V.S. Lien, V. Racapé, R. Castano Primo	S. Tarot

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GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
GLO	Global
MDS	Marine Data Store
MFC	Monitoring and Forecasting Centre
MY	Multi Year
NetCDF	Network Common Data Form
NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
TAC	Thematic Assembly Center

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document describes the reprocessed biogeochemical product INSITU_GLO_BGC_DISCRETE_MY_013_046 distributed by the Copernicus Marine Service In Situ Thematic Assembly Centre (In Situ TAC): how it is built, what is the content, what data services are available to access them, and how to use the files. The product is global and covers the period from 1st January 1990 up to real-time less than eleven months, and it contains five datasets with parameters on chlorophyll-a concentration, oxygen, nutrients (nitrate, silicate, phosphate) and inorganic carbon (dissolved inorganic carbon, total alkalinity)

The In Situ TAC is a distributed system built on the existing activities and services developed within EU-funded projects and support from EuroGOOS Regional alliances. It provides a research and operational framework to develop and deliver in situ observations and derived products to address progressively global and regional needs for monitoring, modeling and downstream service development.

The main objective of the Copernicus Marine Service is to deliver and operate a rigorous, robust and sustainable Ocean Monitoring and Forecasting system to users of all marine applications: maritime safety, marine resources, marine and coastal environment and climate, seasonal and weather forecasting. The In Situ TAC prepares reprocessed datasets for reanalysis activities performed by the Copernicus Marine Service forecasting centers (MFCs) and external users.
The in situ reprocessed biogeochemical product integrates quality flags from the NRT multiparameter product INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030. The quality flags have been reassessed in delayed mode using updated real-time and delayed mode quality assessment procedures for the most common biogeochemical parameters, found in CMEMS-INS-QUID-013-046 (Jaccard et al., 2024).

This product is designed for users that need aggregated datasets with a well-documented and transparent quality control (e.g., ocean forecasting centres, research communities, etc.).

The product is published on the Copernicus Marine Service dissemination server after automatic and human quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files are in NetCDF format.

The reprocessing system is described in the Quality Information Document (QUID): CMEMS-INS-QUID-013-046 (Jaccard et al., 2024).

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	INSITU_GLO_BGC_DISCRETE_MY_013_046
Geographical coverage	Global : 180°W-180°E ; 89°S – 90°N
Variables	• Chl-a fluoresence • Chl-a concentration • Dissolved oxygen • Nitrate • Silicate • Phosphate • Dissolved inorganic carbon • Total alkalinity
Product Type	In Situ Observations
Available time series	1st January 1990 and regularly updated see product improvements pages
Temporal resolution	Observation dependant
Update Frequency	The dataset is updated two times per year: in June when the dataset is extended to include all data up until the end of June the previous year, and in November when the full dataset up until the end of December the previous year is reprocessed.
Target delivery time	N/A
Horizontal resolution	N/A
Number of vertical levels	1 m vertical resolution
Format	NetCDF CF1.11

b) Production System Description

The reprocessed biogeochemical product is based on the NRT product (INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030) history directory files from the Global Production Unit operated by IFREMER. Quality flags for the most relevant parameters have been reprocessed using updated quality assessment procedures. This work has been performed for all platforms for which a sufficient measurement volume was provided.

Quality control procedures were developed and tested by the partners of the Copernicus Marine Service In Situ TAC phase 1 Task 3.2 Methods to assess multi-year historical biogeochemical products during period 2015-2018. More information on these procedures and quality assessments of the product can be found in CMEMS-INS-QUID-013-046 (Jaccard et al., 2024).

- Chl-a Fluorescence

Measurements of chl-a fluorescence are saved in variable FLU2. For historical reasons, variable CPHL is in use for BioArgo platforms. This is the only exception. A variable attribute named sensing_method has been added and set to fluorescence for these files. We hope hereby to avoid the confusion of proxy measurements based on fluorescence to be regarded as concentration based on laboratory analysis from water samples.

- Dissolved Oxygen

Dissolved oxygen in sea water comes in different and original units (as received, depending on the sensor used. This is the reason why product files may have variables DOXY, DOX1 or DOX2.

- Nutrients

Nutrients include nitrate (NTRA and NTAW), silicate (SLCA and SLCW) and phosphate (PHOS and PHOW).

- Carbon

The Carbon dataset contains a subset of the inorganic carbon data distributed in the NRT product, which includes a selection of EMODNet Chemistry acidification data. Variables included are dissolved inorganic carbon (TICW) and total alkalinity (ALKW) in the originally reported units.

- Easy BGC

The Easy BGC dataset includes the easy variables in both µmol kg-1 and mmol m-3 for oxygen concentration (DOXY and DOX2) and nutrients (NTRA, NTAW, SLCA, SLCW, PHOS and PHOW) and in mg m-3 for chlorophyll (CPHL or FLU2 or CHLT), the vertical reference in both dbars and meters (PRES and DEPTH) and the <PARAM>_CONVERSION_METHOD.
<PARAM>_CONVERSION_METHOD details action done for each conversion using a unique code summarized in tables 1 to 3.
The two global attributes “history” and “comment” are enriched with easy BGC version and information

Table 1: unit conversion methods of vertical reference (sea water pressure or depth)

Code	Meaning	Comment
0	no_conversion_needed	no conversion performed
1	conversion_failed	no conversion is performed if "latitude" is missing
2	pres=swdensity0*g*deph*0.0001_from_unesco_1983	-
3	pres(deph,lat)_from_unesco_1983	-
4	deph=pres/(swdensity0*g*0.0001)_from_unesco_1983	-
5	deph(pres,lat)_from_unesco_1983	-

Table 2: unit conversion methods of dissolved oxygen

Code	Meaning	Comment
0	no_conversion_needed	no conversion performed
1	conversion_failed	one useful variable is missing for conversion
2	ml_per_l_to_millimole_per_m3_following_SCOR_WG_142	-
3	micromole_per_kg_to_millimole_per_m3_following_SCOR_WG_142	-
4	ml_per_l_to_micromole_per_kg_following_SCOR_WG_142	_
5	millimole_per_m3_to_micromole_per_kg_following_SCOR_WG_142	-

Table 3: unit conversion methods of nutrients (nitrate, silicate, phosphate)

Code	Meaning	Comment
0	no_conversion_needed	no conversion performed
1	conversion_failed	one useful variable is missing for conversion
2	micromole_per_kg_to_millimole_per_m3_using_sea_water_average density_of_1.025_kg_per_l	For chemical measurements according to Jiang et al. Font. Mar. Sci. 2022
3	micromole_per_kg_to_millimole_per_m3_using_teos10_potenial_density_of_seawater	For sensor measurements according to Jiang et al. Font. Mar. Sci. 2022
4	millimole_per_m3_to_micromole_per_kg_using_sea_water_average density_of_1.025_kg_per_l	For chemical measurements according to Jiang et al. Font. Mar. Sci. 2022
5	millimole_per_m3_to_micromole_per_kg_using_teos10_potenial_density_of_seawater	For sensor measurements according to Jiang et al. Font. Mar. Sci. 2022

QC Procedures

Quality control procedures applied to data in this product have been developed and tested by the partners of Copernicus Marine Service In Situ TAC phase 1, Task 3.2 Methods to assess multi-year historical biogeochemical products during period 2015-2018, and further developed by the Task 6.2 task team. This includes both real-time applicable tests, delayed mode tests as well as some visual controls. Finally, an evaluation against ocean colour products has been applied on the quality flags of in-situ chl-a fluorescence measurements.

More information on these procedures is available in CMEMS-INS-QUID-013-046 (Jaccard et al., 2024).

c) Dataset details

Product Lines	INSITU_GLO_BGC_DISCRETE_MY_013_046	
Geographical coverage	Global	
Datasets: cmems_obs-ins_glo_bgc-chl_my_na_irr cmems_obs-ins_glo_bgc-ox_my_na_irr cmems_obs-ins_glo_bgc-nut_my_na_irr cmems_obs-ins_glo_bgc_my_easybgc_irr cmems_obs-ins_glo_bgc-car_my_na_irr		
Product Type	In Situ Observations	
Update frequency	2 times per year (June and November)	
Available time series	up to 1 year before	
Target delivery time		
Temporal resolution	Observation dependant	

Dataset : cmems_obs-ins_glo_bgc-chl_my_na_irr Variables name in the NetCDF file and Unit: Long_name & Standard_name
CHLT [mg m-3] Total chlorophyll mass_concentration_of_chlorophyll_in_sea_water
CPHL [mg m-3] Chlorophyll-a mass_concentration_of_chlorophyll_a_in_sea_water
FLU2 [mg m-3] Chlorophyll-a fluorescence mass_concentration_of_chlorophyll_a_fluorescence_in_sea_water
Dataset : cmems_obs-ins_glo_bgc-ox_my_na_irr Variables name in the NetCDF file and Unit: Long_name & Standard_name
DOXY [mmol m-3] Dissolved oxygen mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
DOX1 [ml l-1] Dissolved oxygen volume_fraction_of_oxygen_in_sea_water
DOX2 [mmol kg-1] Dissolved oxygen moles_of_oxygen_per_unit_mass_in_sea_water
Dataset : cmems_obs-ins_glo_bgc-nut_my_na_irr Variables name in the NetCDF file and Unit: Long_name & Standard_name
NTRA [mmol m-3] Nitrate (NO3-N) mole_concentration_of_nitrate_in_sea_water
NTAW [µmol kg-1] Nitrate (NO3-N) moles_of_nitrate_per_unit_mass_in_sea_water
SLCA [mmol m-3] Silicate (SIO4-SI) mole_concentration_of_silicate_in_sea_water
SLCW [µmol kg-1]

Silicate (SIO4-SI) moles_of_silicate_per_unit_mass_in_sea_water
PHOS [mmol m-3] Phosphate (PO4-P) mole_concentration_of_phosphate_in_sea_water
PHOW [µmol kg-1] Phosphate (PO4-P) moles_of_phosphate_per_unit_mass_in_sea_water
Dataset : cmems_obs-ins_glo_bgc_my_easybgc_irr Variables name in the NetCDF file and Unit: Long_name & Standard_name
CHLT [mg m-3] Total chlorophyll mass_concentration_of_chlorophyll_in_sea_water
CPHL [mg m-3] Chlorophyll-a mass_concentration_of_chlorophyll_a_in_sea_water
FLU2 [mg m-3] Chlorophyll-a fluorescence mass_concentration_of_chlorophyll_a_fluorescence_in_sea_water
DOXY [mmol m-3] Dissolved oxygen mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
DOX2 [mmol kg-1] Dissolved oxygen moles_of_oxygen_per_unit_mass_in_sea_water
NTRA [mmol m-3] Nitrate (NO3-N) mole_concentration_of_nitrate_in_sea_water
NTAW [µmol kg-1] Nitrate (NO3-N) moles_of_nitrate_per_unit_mass_in_sea_water
SLCA [mmol m-3] Silicate (SIO4-SI) mole_concentration_of_silicate_in_sea_water
SLCW [µmol kg-1] Silicate (SIO4-SI) moles_of_silicate_per_unit_mass_in_sea_water
PHOS [mmol m-3] Phosphate (PO4-P) mole_concentration_of_phosphate_in_sea_water
PHOW [µmol kg-1] Phosphate (PO4-P) moles_of_phosphate_per_unit_mass_in_sea_water
Dataset : cmems_obs-ins_glo_bgc-car_my_na_irr Variables name in the NetCDF file and Unit: Long_name & Standard_name
TICW [µmol kg-1] Dissolved inorganic carbon moles_of_dissolved_inorganic_carbon_per_unit_mass_in_sea_ water
ALKW [µmol kg-1] Total alkalinity

sea_water_alkalinity_per_unit_mass_expressed_as_mole_equivalent

d) Additional information on some parameters

The missing value for this product is: -2147483647

Land and sea-ice masks are equal to "_FillValue" (see variable attribute on NetCDF file).

e) Processing information

The product is updated two times per year.
The product covers the period from 1st January 1990 up to launch date less eleven months

f) Catalogue updating

Not applicable.

g) Time averaging

The fields are instantaneous.

h) Product use and impact

The BGC and Carbon data products play an important role for monitoring and understanding the health of marine ecosystems, assessing the impacts of climate change, and supporting policy and sustainable management. The BGC data products can be used for data assimilation, research requiring high-resolution, long-term datasets, and validation of models and satellite observations. Compared to other products, the BGC data products have large value for decadal variability and trend analysis in biogeochemical processes and ecosystem dynamics, contributing to environmental monitoring and climate impact assessments.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1 FILE SIZE

The size of each file depends on the type of instrument and the length of the series of data. It varies from about 60 Kb to about 230 Mb.

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex,** for each dataset.

5) REFERENCES

Jaccard, P. et al. (2024) Quality Information Document In Situ TAC INSITU_GLO_BGC_DISCRETE_MY_013_046. And Synthetic Quality Overview document (SQO). https://doi.org/10.13155/54846

Jiang, L.-Q., Pierrot, D., Wanninkhof, R., Feely, R. A., Tilbrook, B., Alin, S., Barbero, L., Byrne, R. H., Carter, B. R., Dickson, A. G., Gattuso, J.-P., Greeley, D., Hoppema, M., Humphreys, M. P., Karstensen, J., Lange, N., Lauvset, S. K., Lewis, E. R., Olsen, A., Pérez, F. F., Sabine, C., Sharp, J. D., Tanhua, T., Trull, T. W., Velo, A., Allegra, A. J., Barker, P., Burger, E., Cai, W.-J., Chen, C.-T. A., Cross, J., Garcia, H., Hernandez-Ayon, J. M., Hu, X., Kozyr, A., Langdon, C., Lee, K., Salisbury, J., Wang, Z. A., and Xue, L.: Best Practice Data Standards for Discrete Chemical Oceanographic Observations, Front. Mar. Sci., 8, https://doi.org/10.3389/fmars.2021.705638, 2022.

UNESCO 1983. Algorithms for computation of fundamental properties of seawater, (1983). UNESCO Tech. Pap. in Mar. Sci., No. 44, 53 pp.

SCOR WG 142, Bittig, H. C., Körtzinger, A., Johnson, K. S., Claustre, H., Emerson, S., Fennel, K., et al. (2016). Quality Control Procedures for Oxygen and Other Biogeochemical Sensors on Floats and Gliders. Recommendations on the Conversion between Oxygen Quantities for Bio-Argo Floats and Other Autonomous Sensor Platforms. Doi: 10.13155/45915

6) ANNEX

Below is an example of an output file showing all information provided in the files.
File "GL_PR_BO_18DD9201.nc"
File type: Hierarchical Data Format, version 5

```
netcdf BGC_REP_2024NOV/ARC/BO/GL_PR_BO_18DD9201.nc {
  dimensions:
    DEPTH = 7;
    TIME = 4;
    STRING32 = 32;
    STRLEN = 8;
  variables:
    double TIME(TIME=4);
      :long_name = "Time";
      :standard_name = "time";
      :units = "days since 1950-01-01T00:00:00Z";
      :valid_min = -90000.0; // double
      :valid_max = 90000.0; // double
      :uncertainty = " ";
      :axis = "T";
      :ancillary_variables = "TIME_QC";
      :calendar = "standard";
      :_ChunkSizes = 4U; // uint

    byte TIME_QC(TIME=4);
      :long_name = "Time quality flag";
      :valid_min = 0B; // byte
      :valid_max = 9B; // byte
      :flag_values = 0B, 1B, 2B, 3B, 4B, 5B, 6B, 7B, 8B, 9B; // byte
      :flag_meanings = "no_qc_performed good_data probably_good_data
bad_data_that_are_potentially_correctable bad_data value_changed value_below_detection
nominal_value interpolated_value missing_value";
      :_FillValue = -127B; // byte
      :_ChunkSizes = 4U; // uint

    float LATITUDE(TIME=4);
      :long_name = "Latitude of each location";
      :standard_name = "latitude";
      :units = "degree_north";
      :valid_min = -90.0f; // float
      :valid_max = 90.0f; // float
      :uncertainty = " ";
      :axis = "Y";
      :ancillary_variables = "POSITION_QC";
      :_ChunkSizes = 4U; // uint

    float LONGITUDE(TIME=4);
      :long_name = "Longitude of each location";
      :standard_name = "longitude";
      :units = "degree_east";
      :valid_min = -180.0f; // float
```

```
      :valid_max = 180.0f; // float
      :uncertainty = " ";
      :axis = "X";
      :ancillary_variables = "POSITION_QC";
      :_ChunkSizes = 4U; // uint

    byte POSITION_QC(TIME=4);
      :long_name = "Position quality flag";
      :valid_min = 0B; // byte
      :valid_max = 9B; // byte
      :flag_values = 0B, 1B, 2B, 3B, 4B, 5B, 6B, 7B, 8B, 9B; // byte
      :flag_meanings = "no_qc_performed good_data probably_good_data
bad_data_that_are_potentially_correctable bad_data value_changed value_below_detection
nominal_value interpolated_value missing_value";
      :_FillValue = -127B; // byte
      :_ChunkSizes = 4U; // uint

    char DC_REFERENCE(TIME=4, STRING32=32);
      :long_name = "Station/Location unique identifier in data centre";
      :_FillValue = " ";
      :_ChunkSizes = 4U, 32U; // uint

    char DIRECTION(TIME=4);
      :long_name = "Direction of the profiles";
      :flag_values = "A, D, U";
      :flag_meanings = "ascending_profile descending_profile unknown";
      :_FillValue = " ";
      :_ChunkSizes = 4U; // uint

    char TRAJECTORY(STRLEN=8);
      :long_name = "trajectory";
      :cf_role = "trajectory_id";
      :_ChunkSizes = 8U; // uint

    float DEPH(TIME=4, DEPTH=7);
      :_FillValue = 9.96921E36f; // float
      :long_name = "Depth";
      :standard_name = "depth";
      :units = "m";
      :valid_min = -12000.0f; // float
      :valid_max = 12000.0f; // float
      :uncertainty = " ";
      :axis = "Z";
      :positive = "down";
      :reference = "sea_level";
      :data_mode = "R";
      :ancillary_variables = "DEPH_QC";
      :_ChunkSizes = 4U, 7U; // uint

    byte DEPH_QC(TIME=4, DEPTH=7);
      :valid_max = 9B; // byte
      :flag_values = 0B, 1B, 2B, 3B, 4B, 5B, 6B, 7B, 8B, 9B; // byte
      :_FillValue = -127B; // byte
      :long_name = "Depth quality flag";
```

```
      :valid_min = 0B; // byte
      :flag_meanings = "no_qc_performed good_data probably_good_data
bad_data_that_are_potentially_correctable bad_data value_changed value_below_detection
nominal_value interpolated_value missing_value";
      :_ChunkSizes = 4U, 7U; // uint

    float DOX2(TIME=4, DEPTH=7);
      :long_name = "Dissolved oxygen";
      :data_mode = "R";
      :ancillary_variables = "DOX2_QC";
      :standard_name = "moles_of_oxygen_per_unit_mass_in_sea_water";
      :units = "µmol kg-1";
      :coordinates = "TIME LATITUDE LONGITUDE DEPH TRAJECTORY";
      :_FillValue = 9.96921E36f; // float
      :_ChunkSizes = 4U, 7U; // uint

    byte DOX2_QC(TIME=4, DEPTH=7);
      :valid_max = 9B; // byte
      :flag_values = 0B, 1B, 2B, 3B, 4B, 5B, 6B, 7B, 8B, 9B; // byte
      :flag_meanings = "no_qc_performed good_data probably_good_data
bad_data_that_are_potentially_correctable bad_data value_changed value_below_detection
nominal_value interpolated_value missing_value";
      :coordinates = "TIME LATITUDE LONGITUDE DEPH TRAJECTORY";
      :long_name = "Dissolved oxygen quality flag";
      :valid_min = 0B; // byte
      :_FillValue = -127B; // byte
      :_ChunkSizes = 4U, 7U; // uint

  // global attributes:
  :platform_code = "18DD9201";
  :platform_name = "UNKNOWN";
  :wmo_platform_code = "18DD9201";
  :ices_platform_code = " ";
  :coriolis_platform_code = "18DD9201";
  :site_code = " ";
  :comment = " ";
  :source = "unknown";
  :source_platform_category_code = "0";
  :institution = "Unknown institution";
  :institution_edmo_code = "1051";
  :institution_country = "Unknown";
  :institution_references = " ";
  :naming_authority = "Copernicus Marine In Situ";
  :title = "Global Ocean - In Situ Observation Copernicus";
  :summary = " ";
  :data_mode = "R";
  :wmo_instrument_type = 810; // int
  :area = "Global Ocean";
  :geospatial_lat_min = "48.53580";
  :geospatial_lat_max = "48.84430";
  :geospatial_lon_min = "-125.45420";
  :geospatial_lon_max = "-125.03430";
  :geospatial_vertical_min = "1.00";
  :geospatial_vertical_max = "90.00";
```

:time_coverage_start = "1992-10-29T09:15:00Z";
:time_coverage_end = "1992-10-31T02:09:00Z";
:cdm_data_type = "profile";
:featureType = "trajectoryProfile";
:bottom_depth = " ";
:last_date_observation = "1992-10-31T02:09:00Z";
:last_latitude_observation = "48.64980";
:last_longitude_observation = "-125.03430";
:references = "http://marine.copernicus.eu http://www.marineinsitu.eu";
:license = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence";
:update_interval = "PT1H";
:citation = "These data were collated within the Copernicus Marine Service (In Situ) and EMODnet collaboration framework. Data is made freely available by the Copernicus Marine Service and the programs that contribute to it.";
:doi = "https://doi.org/10.13155/59938 https://doi.org/10.13155/40846 https://doi.org/10.13155/53381 https://doi.org/10.13155/36230 https://doi.org/10.13155/43494";
:creator_name = " ";
:id = "GL_PR_BO_18DD9201";
:netcdf_version = "netCDF-4 classic model";
:format_version = "2.0";
:Conventions = "CF-1.11 Copernicus-InSituTAC-FormatManual-2.0.0 Copernicus-InSituTAC-ParametersList-3.3.0 Copernicus-InSituTAC-AttributesList-1.0.0";
:date_modified = "2023-11-21T19:56:16Z";
:history = "2023-11-21T19:56:16Z : Creation";
:publisher_email = "cmems-service@ifremer.fr";
:publisher_name = "Copernicus Marine Service";
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